

can become tiresome when done in the straightforward way. Fortunately, there is a faster method. Define

$$\begin{aligned} p_{-2} &= 0, & p_{-1} &= 1, & q_{-2} &= 1, & q_{-1} &= 0, \\ p_{n+1} &= a_{n+1}p_n + p_{n-1} \\ q_{n+1} &= a_{n+1}q_n + q_{n-1}. \end{aligned}$$

Then

$$\frac{p_n}{q_n} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_n}}}}.$$

Using these relations, we can compute the partial quotients p_n/q_n from the previous ones, rather than having to start a new computation every time a new a_n is found.

3.13 Exercises

1. (a) Find integers x and y such that $17x + 101y = 1$.
(b) Find $17^{-1} \pmod{101}$.
2. (a) Solve $7d \equiv 1 \pmod{30}$.
(b) Suppose you write a message as a number $m \pmod{31}$. Encrypt m as $m^7 \pmod{31}$. How would you decrypt? (*Hint*: Decryption is done by raising the ciphertext to a power mod 31. Fermat's theorem will be useful.)
3. (a) Find all solutions of $12x \equiv 28 \pmod{236}$.
(b) Find all solutions of $12x \equiv 30 \pmod{236}$.
4. (a) Use the Euclidean algorithm to compute $\gcd(30030, 257)$.
(b) Using the result of part (a) and the fact that $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, show that 257 is prime. (*Remark*: This method of computing one gcd, rather than doing several trial divisions (by 2, 3, 5, ...), is often faster for checking whether small primes divide a number.)
5. (a) Compute $\gcd(4883, 4369)$.
(b) Factor 4883 and 4369 into products of primes.
6. (a) Let $F_1 = 1, F_2 = 1, F_{n+1} = F_n + F_{n-1}$ define the Fibonacci numbers 1, 1, 2, 3, 5, 8, ... Use the Euclidean algorithm to compute

- (b) Find $\gcd(11111111, 11111)$.
- (c) Let $a = 111 \cdots 11$ be formed with F_n repeated 1's and let $b = 111 \cdots 11$ be formed with F_{n-1} repeated 1's. Find $\gcd(a, b)$. (*Hint*: Compare your computations in parts (a) and (b).)
7. (a) Let p be prime. Suppose a and b are integers such that $ab \equiv 0 \pmod{p}$. Show that either $a \equiv 0$ or $b \equiv 0 \pmod{p}$.
(b) Show that if a, b, n are integers with $n|ab$ and $\gcd(a, n) = 1$, then $n|b$.
8. Let $p \geq 3$ be prime. Show that the only solutions to $x^2 \equiv 1 \pmod{p}$ are $x \equiv \pm 1 \pmod{p}$. (*Hint*: Apply Exercise 7(a) to $(x+1)(x-1)$.)
9. Suppose $x \equiv 2 \pmod{7}$ and $x \equiv 3 \pmod{10}$. What is x congruent to mod 70?
10. A group of people are arranging themselves for a parade. If they line up three to a row, one person is left over. If they line up four to a row, two people are left over, and if they line up five to a row, three people are left over. What is the smallest possible number of people? What is the next smallest number? (*Hint*: Interpret this problem in terms of the Chinese remainder theorem.)
11. Let p be prime. Show that $a^p \equiv a \pmod{p}$ for all a .
12. Divide 2^{10203} by 101. What is the remainder?
13. Find the last 2 digits of 123^{562} .
14. (a) Evaluate $7^7 \pmod{4}$.
(b) Use part (a) to find the last digit of 7^{7^7} . (*Note*: a^{bc} means $a^{(bc)}$ since the other possible interpretation would be $(a^b)^c = a^{bc}$, which is written more easily without a second exponentiation.)
15. (a) Compute $\phi(d)$ for all of the divisors of 10 (namely, 1, 2, 5, 10), and find the sum of these $\phi(d)$.
(b) Repeat part (a) for all of the divisors of 12.
(c) Let $n \geq 1$. Conjecture the value of $\sum \phi(d)$, where the sum is over the divisors of n . (This result is proved in many elementary number theory texts.)
16. (a) Let $p = 7, 13$, or 19. Show that $a^{1728} \equiv 1 \pmod{p}$ for all a with $p \nmid a$.

- (b) Let $p = 7, 13$, or 19 . Show that $a^{1729} \equiv a \pmod{p}$ for all a . (*Hint*: Consider the case $p|a$ separately.)
- (c) Show that $a^{1729} \equiv a \pmod{1729}$ for all a . Composite numbers n such that $a^n \equiv a \pmod{n}$ for all a are called Carmichael numbers. They are rare (561 is another example), but there are infinitely many of them [Alford et al. 2].
17. (a) Show that every nonzero congruence class mod 11 is a power of 2, and therefore 2 is a primitive root mod 11.
- (b) Note that $2^3 \equiv 8 \pmod{11}$. Find x such that $8^x \equiv 2 \pmod{11}$. (*Hint*: What is the inverse of 3 (mod 10)?)
- (c) Show that every nonzero congruence class mod 11 is a power of 8, and therefore 8 is a primitive root mod 11.
- (d) Let p be prime and let g be a primitive root mod p . Let $h \equiv g^y \pmod{p}$ with $\gcd(y, p-1) = 1$. Let $xy \equiv 1 \pmod{p-1}$. Show that $h^x \equiv g \pmod{p}$.
- (e) Let p and h be as in part (d). Show that h is a primitive root mod p . (*Remark*: Since there are $\phi(p-1)$ possibilities for the exponent x in part (d), this yields all of the $\phi(p-1)$ primitive roots mod p .)
18. (a) Find the inverse of $\begin{pmatrix} 1 & 1 \\ 6 & 1 \end{pmatrix} \pmod{26}$.
- (b) Find all values of $b \pmod{26}$ such that $\begin{pmatrix} 1 & 1 \\ b & 1 \end{pmatrix} \pmod{26}$ is invertible.
19. Find all primes p for which $\begin{pmatrix} 3 & 5 \\ 7 & 3 \end{pmatrix} \pmod{p}$ is not invertible.
20. Let a and $n > 1$ be integers with $\gcd(a, n) = 1$. The **order** of a mod n is the smallest positive integer r such that $a^r \equiv 1 \pmod{n}$. We denote $r = \text{ord}_n(a)$.
- (a) Show that $r \leq \phi(n)$.
- (b) Show that if $m = rk$ is a multiple of r , then $a^m \equiv 1 \pmod{n}$.
- (c) Suppose $a^t \equiv 1 \pmod{n}$. Write $t = qr + s$ with $0 \leq s < r$ (this is just division with remainder). Show that $a^s \equiv 1 \pmod{n}$.
- (d) Using the definition of r and the fact that $0 \leq s < r$, show that $s = 0$ and therefore $r|t$. This, combined with part (b), yields the result that $a^t \equiv 1 \pmod{n}$ if and only if $\text{ord}_n(a)|t$.

(e) Show that $\text{ord}_n(a)|\phi(n)$.

21. This exercise will show by example how to use the results of Exercise 20 to prove a number is a primitive root mod a prime p , once we know the factorization of $p-1$. In particular, we'll show that 7 is a primitive root mod 601. Note that $600 = 2^3 \cdot 3 \cdot 5^2$.

- (a) Show that if an integer $r < 600$ divides 600, then it divides at least one of 300, 200, 120 (these numbers are $600/2$, $600/3$, and $600/5$).
- (b) Show that if $\text{ord}_{601}(7) < 600$, then it divides one of the numbers 300, 200, 120.
- (c) A calculation shows that

$$7^{300} \equiv 600, \quad 7^{200} \equiv 576, \quad 7^{120} \equiv 423 \pmod{601}.$$

Why can we conclude that $\text{ord}_{601}(7)$ does not divide 300, 200, or 120?

- (d) Show that 7 is a primitive root mod 601.
- (e) In general, suppose p is a prime and $p-1 = q_1^{a_1} \cdots q_s^{a_s}$ is the factorization of $p-1$ into primes. Describe a procedure to check whether a number g is a primitive root mod p . (Therefore, if we need to find a primitive root mod p , we can simply use this procedure to test the numbers $g = 2, 3, 5, 6, \dots$ in succession until we find one that is a primitive root.)

22. We want to find an exponent k such that $3^k \equiv 2 \pmod{65537}$.

- (a) Observe that $2^{32} \equiv 1 \pmod{65537}$, but $2^{16} \not\equiv 1 \pmod{65537}$. It can be shown (Exercise 32) that 3 is a primitive root mod 65537, which implies that $3^n \equiv 1 \pmod{65537}$ if and only if $65536|n$. Use this to show that $2048|k$ but 4096 does not divide k . (*Hint*: Raise both sides of $3^k \equiv 2$ to the 16th and to the 32nd powers.)
- (b) Use the result of part (a) to conclude that there are only 16 possible choices for k that need to be considered. Use this information to determine k . This problem shows that if $p-1$ has a special structure, for example, a power of 2, then this can be used to avoid exhaustive searches. Therefore, such primes are cryptographically weak. See Exercise 9 in Chapter 7 for a reinterpretation of the present problem.

23. (a) Let $x = b_1b_2 \dots b_w$ be an integer written in binary (for example, when $x = 1011$, we have $b_1 = 1, b_2 = 0, b_3 = 1, b_4 = 1$). Let y and n be integers. Perform the following procedure: